

OSI MODEL

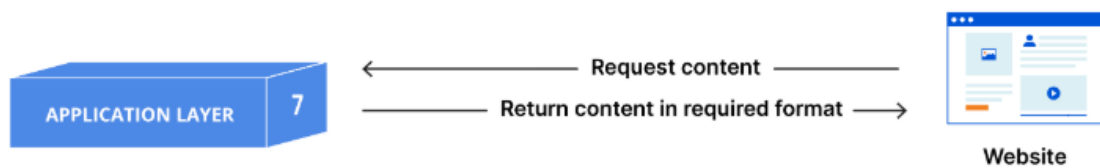
OSI model is reference model can help to understand network & how data packet can travel from source to destination.

it can split communication system into 7 layers-

7. Application Layer- It act as interface between end user application & network.

Application layer provides necessary protocols & services for applications to access and interact with network resources.

Protocol – HTTP, HTTPS, SMTP, IMAP, POP3, TELNET, SSH, FTP

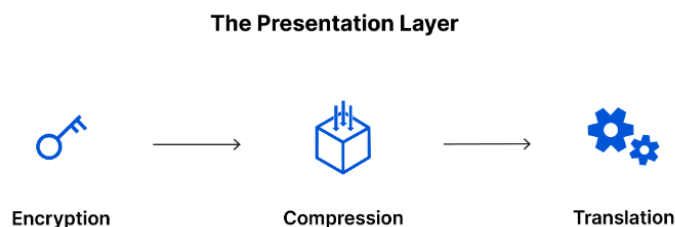


6. Presentation Layer-Represents data in suitable format -image (jpeg, png), video (mp4), audio (mp3).

The presentation layer is responsible for translation, [encryption](#), and compression of data.

Translation (data 1 format to another format without changing meaning)

Compression (process reduce size of data while preserving essential information)-enhance speed, efficiency in communication.



5. Session Layer

This is the layer responsible for opening and closing communication between the two devices. It ensures session stay open during data exchange & close session for avoiding resources waste(Bandwidth, Download resume from last checkpoint).

Protocol –

NetBIOS - It allowing applications on different computers to communicate over LAN.
le- Windows networks, NetBIOS was used for file and printer sharing

RPC (Remote Procedure Call) - RPC allows a program to execute a procedure (function) on another computer as if it were local. In distributed systems, a client might use RPC to request data from a database server. For example, a banking app might use RPC to fetch account details from a remote server.

The Session Layer



Session of communication

4 Transport Layer

TCP - Segmentation -----> Sequencing -----> Retransmission -----> Reassemble

This layer responsible for end-to-end communication between two devices. At sender end transport layer take data from session layer & then break it into chunks (Segments), and then transmit to network layer, after then at receiver end transport layer can assemble these segments into data then flow to session layer.

Transport Layer



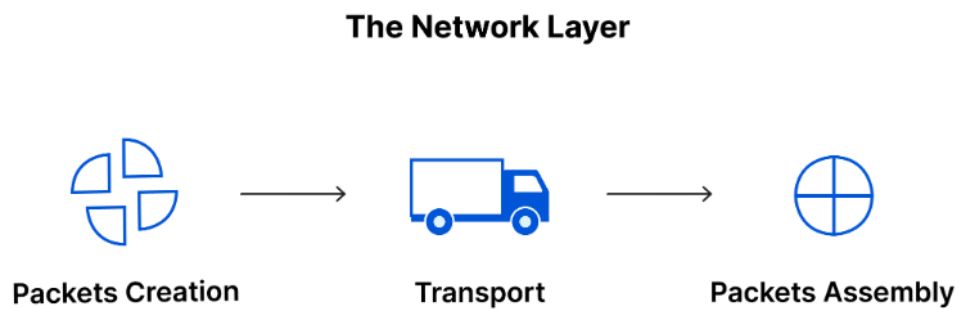
Flow control (optimal speed data transmission from sender to receiver) & Error control (At receiver ensures that complete data is transmit if not then transport layer retransmit data).

Sockets provide a way for programs to communicate over a network using standard protocols like TCP or UDP.

3. Network Layer

The [network layer](#) is responsible for facilitating data transfer between two different networks. The network layer also finds the best physical path for the data to reach its destination; this is known as [routing](#).

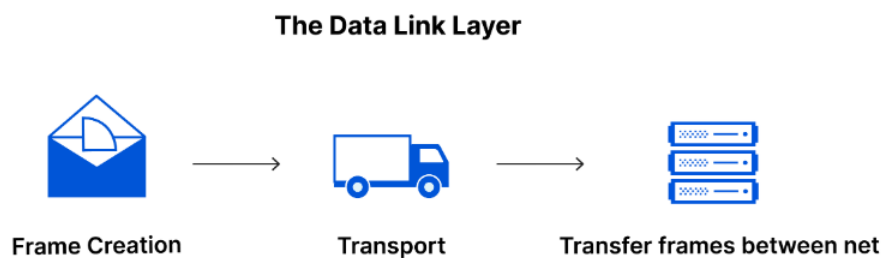
Protocol – ICMP,EIGRP,BGP,OSPF,RIP



2. DLL (Data Link layer)

Data link layer facilitates data transfer between two devices on the *same* network. Data link layer is also responsible for flow control and error control in intra-network communication (The transport layer only does flow control and error control for inter-network communications).

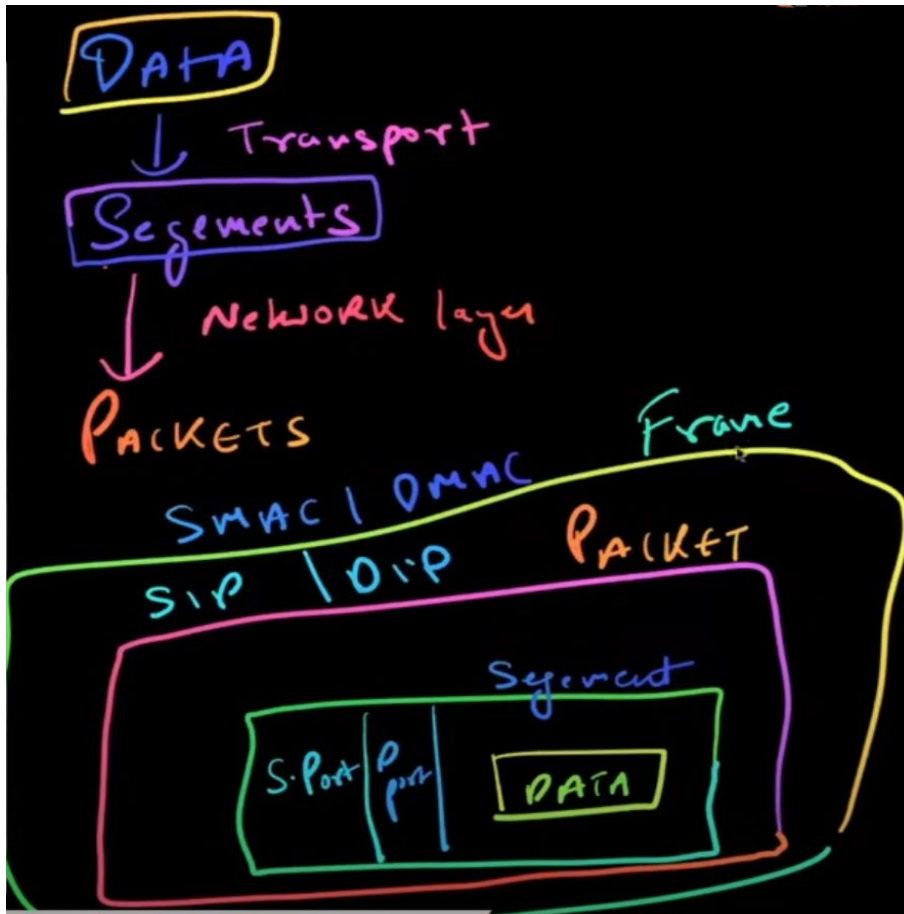
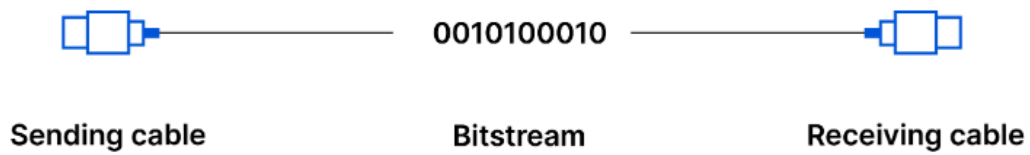
Protocol - Ethernet, PPP, Wi-Fi (802.11).



1. Physical Layer

This layer includes the physical equipment involved in the data transfer, such as the cables and [switches](#).

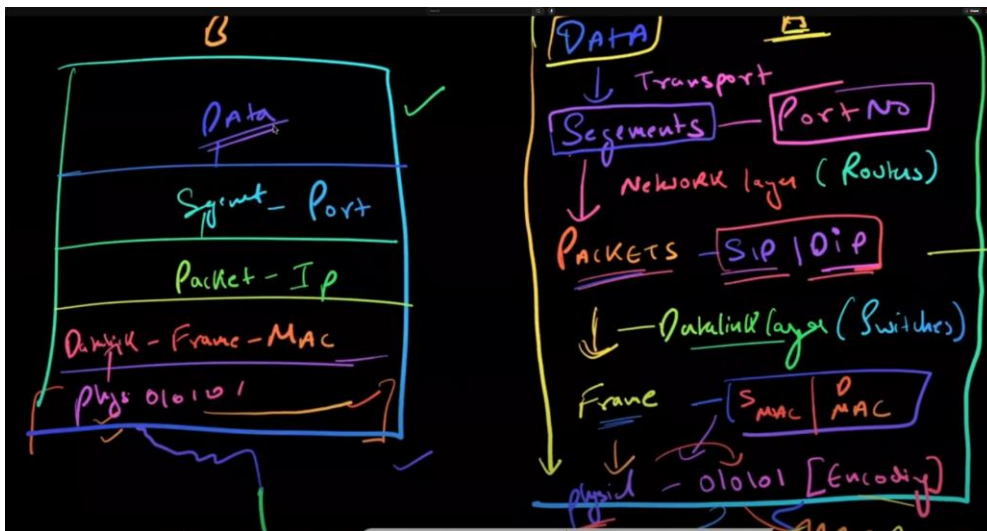
The Physical Layer



Segments – Encapsulate (DATA + Source Port & Destination Port).

Packet – Source IP + Destination IP + (Segments).

Frame - Source MAC + Destination MAC + (Packet).



Layer 7: Application Layer Attacks

SQL Injection – Manipulating database queries.

Cross-Site Scripting (XSS) – Injecting malicious scripts into web pages.

Cross-Site Request Forgery (CSRF) – Forcing a user to perform actions without consent.

DNS Spoofing/Cache Poisoning – Redirecting domain requests to malicious sites.

Phishing– Tricking users into revealing sensitive info.

Malware Injection – Planting malicious code into apps.

Layer 6: Presentation Layer Attacks

SSL/TLS Attacks – Exploiting weak encryption (e.g., POODLE, BEAST).

Man-in-the-Middle (MITM) – Intercepting and modifying encrypted traffic.

Data Serialization Attacks– Injecting malicious payloads into data formats.

Code Injection in File Formats – Exploiting parsing vulnerabilities in PDFs, images, etc

Layer 5: Session Layer Attacks

Session Hijacking – Stealing session tokens or IDs to impersonate a user.

Session Fixation – Forcing a victim to use a known session ID.

RPC Exploits – Attacking Remote Procedure Call services to gain control.

Layer 4: Transport Layer Attacks

SYN Flood Attack– Exploiting TCP handshake to exhaust server resources.

UDP Flood Attack – Overwhelming with high-volume UDP packets.

Port Scanning – Probing ports to find vulnerabilities.

Layer 3: Network Layer Attacks

IP Spoofing – Faking source IP addresses to impersonate systems.

Route Injection – Inserting false routing information.

ICMP Flood / Smurf Attack – Overloading a network with ICMP traffic.

Ping of Death – Sending oversized ICMP packets to crash systems.

DoS/DDoS (Network-based) – Overwhelming network bandwidth/resources.

Layer 2: Data Link Layer Attacks

MAC Flooding – Overloading a switch's MAC table to force it into broadcast mode.

MAC Spoofing – Impersonating another device's MAC address.

ARP Spoofing/Poisoning – Attacker linking MAC address to IP address another host.

Switch Port Stealing – Forcing traffic to be sent to an attacker's port.

Layer 1: Physical Layer Attacks

Cable Tapping – Physically intercepting communication lines to capture data.

Signal Jamming – Disrupting wireless or radio communications with interference.

Hardware Tampering– Modifying or inserting malicious chips/devices.

Power Disruption Attacks – Cutting power or using surges to damage equipment.

Email Transmission Through the OSI Model

When you send an email, it goes through a process of encapsulation and de-encapsulation across the OSI layers.

1. Encapsulation (Sender's Host):

* Layer 7 (Application): Your email client (e.g., Outlook) uses the Simple Mail Transfer Protocol (SMTP) to format the email message.

- * Layer 6 (Presentation): The email's data (text, attachments) is formatted and potentially compressed and encrypted.
- * Layer 5 (Session): A session is established with the email server to ensure the connection remains open long enough for the email to be sent.
- * Layer 4 (Transport): The email data is broken into smaller chunks called segments. A TCP header is added to each segment, including a port number (port 25 for SMTP) to ensure reliable delivery.
- * Layer 3 (Network): Each segment is encapsulated into a packet. An IP header is added, containing the source and destination IP addresses.
- * Layer 2 (Data Link): Packets are further encapsulated into frames. The sender's MAC address and the next hop's (e.g., the local router's) MAC address are added to the frame header.
- * Layer 1 (Physical): The frames are converted into a stream of bits and sent over the physical medium (e.g., Ethernet cable, Wi-Fi).

2. Transmission Across the Network:

The bits travel across the network, passing through various devices (switches, routers) that operate on different OSI layers. Each device reads the header information at its respective layer to forward the data to the next hop until it reaches the destination network.

3. De-encapsulation (Receiver's Host):

- * Layer 1 (Physical): The receiving host's network card receives the bitstream.
- * Layer 2 (Data Link): The bits are reassembled into frames, and the MAC address is verified.
- * Layer 3 (Network): The frames are converted back into packets, and the IP address is checked to confirm the data is for this host.
- * Layer 4 (Transport): The packets are reassembled into segments, and the TCP header is used to ensure they are in the correct order and re-request any missing segments.
- * Layer 5 (Session): The session is managed to handle the incoming data.
- * Layer 6 (Presentation): The data is decrypted and decompressed back into its original format.
- * Layer 7 (Application): The data is presented to the recipient's email client, where they can read the email.

Network Devices by OSI Layer

- * Layer 1 (Physical): Hubs and Repeaters. Hubs broadcast data to all connected devices, and repeaters amplify signals to extend network range. They don't interpret data.
- * Layer 2 (Data Link): Switches and Bridges. These devices use MAC addresses to forward data frames to the correct destination within a local network, reducing unnecessary traffic.

* Layer 3 (Network): Routers and Layer 3 Switches. Routers are the primary devices at this layer. They use IP addresses to intelligently forward packets between different networks (e.g., between your home network and the internet).

* Layer 4–7 (Transport to Application): Gateways and Firewalls. These devices can operate at multiple layers, often inspecting and filtering traffic based on port numbers (Layer 4) and even application-specific protocols (Layer 7) to enforce security policies.

Attribute	EIGRP	BGP	OSPF	RIP
Full Form	Enhanced Interior Gateway Routing Protocol	Border Gateway Protocol	Open Shortest Path First	Routing Information Protocol
Type	Hybrid (Distance Vector + Link-State features)	Path Vector	Link-State	Distance Vector
Layer	Network Layer	Application Layer	Network Layer	Network Layer
Routing Type	Interior Gateway Protocol (IGP)	Exterior Gateway Protocol (EGP)	Interior Gateway Protocol (IGP)	Interior Gateway Protocol (IGP)
Metric	Bandwidth, Delay	Path Attributes (AS-Path, etc.)	Cost (based on bandwidth)	Hop Count (max 15)
Convergence Speed	Fast	Slow	Fast	Slow
Administrative Distance	90 (internal), 170 (external)	20 (external), 200 (internal)	110	120
Use Case	Cisco-based enterprise networks	Internet routing between ISPs	Large enterprise networks	Small/simple networks

